

Digital Electronics Laboratory

Laboratory Exercise 4. Flip Flops

Objectives:

- To study the fundamentals of basic memory units
- To become familiar with various types of flip-flops
- To implement a data register.

Introduction

In this laboratory we will build on concepts that we examined in the previous laboratory sessions. We are now examining flip-flops. It is likely that we have not examined flip-flops in lectures at this point, but hopefully the laboratory exercise will help explain the concepts as we go along. This laboratory will examine your interpretation of what is occurring after you connect up the circuits.

Pre-Laboratory

The task that you must perform **prior** to sitting this laboratory:

- Read the laboratory assignment in full.

The Laboratory:

Section 1. The Asynchronous RS Flip-Flop

(a) Connect the two NOR gates as shown in Figure 1.

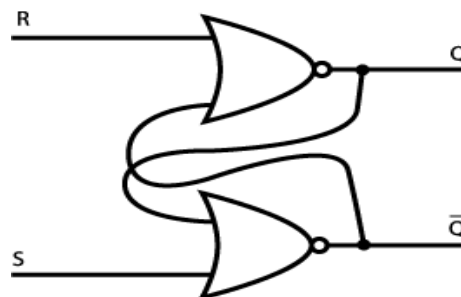


Figure 1. An RS Flip-Flop created using NOR gates.

(b) Vary the inputs R and S (i.e. 0 and 1) to obtain all the possible combinations for Q and $\bar{Q}$.

(c) In your opinion why is there a Q and /Q output? In your opinion how does this circuit work?

Section 2. The Synchronous Flip-Flop

(a) Synchronous means that this flip-flop is concerned with time! Digital circuits can have a concept of time using a clock signal. The clock signal simply goes from low-to-high and high-to-low in a short period of time.

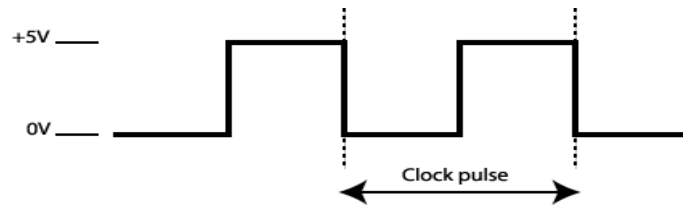


Figure 2. A typical clock signal.

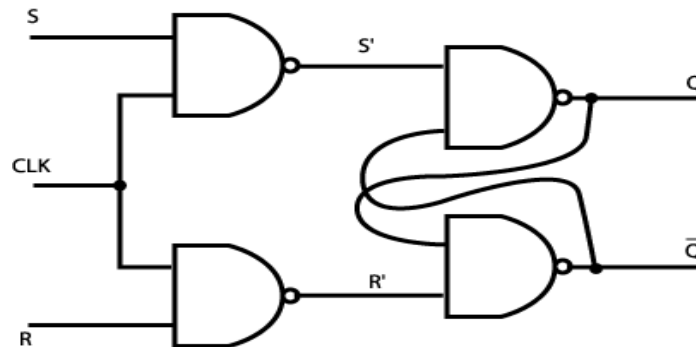


Figure 3. The Synchronous Flip Flop.

Implement the circuit in Figure 3. You can simulate a clock signal by moving the clock line from low to high and back again to low.

(b) Vary inputs R and S and apply the clock pulse. Write the output states into a table as below:

Q _n /Q _n	R	S	Q _{n+1} /Q _{n+1}
0 1	0 0		
1 0	0 0		
0 1	0 1		
1 0	0 1		
0 1	1 0		
1 0	1 0		
0 1	1 1		
1 0	1 1		

(c) Convert the circuit into a D-type flip flop (as shown in Figure 4.)

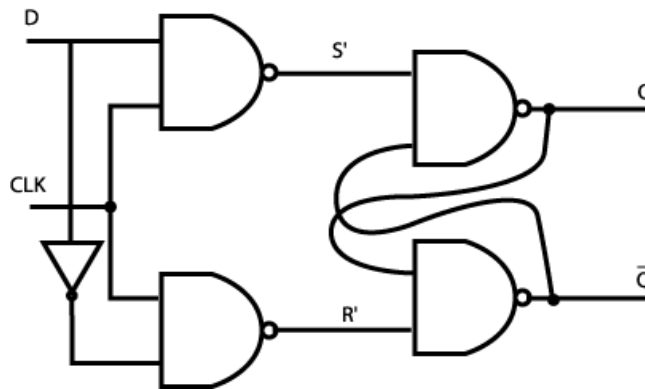


Figure 4. The D-type flip-flop

Draw up the truth table for a D-type flip flop. In your opinion how does it work? what could this circuit be useful for?

Section 3. The J-K Flip-Flop

(a) Connect one of flip flops as shown in Figure 5.

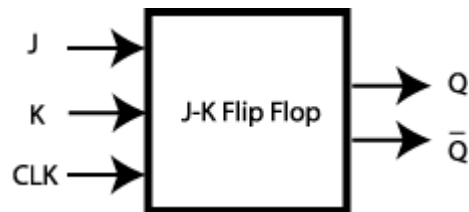


Figure 5. The JK flip-flop

Examine the operation of this flip-flop giving its outputs.

Section 4. Conclusions

State briefly, but clearly, what you have gained from this laboratory.